

T.E./Insem.-501**T.E. (Civil)****HYDROLOGY AND WATER RESOURCES ENGINEERING****(2012 Pattern) (Semester - I)****Time : 1 Hour]****[Max. Marks : 30****Instructions to the candidates:**

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

UNIT - I**Q1) a) Briefly explain rainfall intensity duration frequency curves. [4]**

- b) Calculate the optimum number of rain gauge stations for the given catchment with following data. Assume the error in the rainfall estimation as 70%. [6]

Station Number	1	2	3	4
Rainfall in mm	700	500	300	440

OR**Q2) a) Calculate the value of ϕ -index from the following data of storm of 8cm precipitation that resulted in a direct run-off of 4.4 cm. [6]**

Time in Hours	1	2	3	4	5	6
Incremental rainfall per hour in cm	0.57	0.58	1.25	3.00	1.4	1.2

- b) State the various methods of determination of the mean precipitation over a given catchment area. Explain giving neat sketches any one method. [4]

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UNIT - II

- Q3)** a) Define irrigation and state its necessity in our country. [4]
b) Explain the meaning of consumptive use of water and state the factors on which the consumptive use depends. [6]

OR

- Q4)** a) What are the various methods of assessment of canal revenue? Explain any one. [4]
b) Calculate the discharge required at the head of the distributary from the following: [6]
i) Gross command area of distributary = 12000 hectares.
ii) Culturable irrigable area = 80% of gross area.
iii) Intensity of irrigation = 50% for Rabi.
iv) Intensity of irrigation = 25% for Kharif.
v) Average duty at the head of distributary = 2000 hect/cumecs for Rabi.
vi) Average duty at the head of distributary = 900 hect/cumecs for Kharif.

UNIT - III

- Q5)** a) Define (any 5): [5]
i) Aquifer
ii) Aquiclude
iii) Specific Yield
iv) Specific retention
v) Aquitard
vi) Coefficient of Transmissibility
b) List the various types of tube well. Explain any one with neat sketch. [5]

OR

- Q6)** a) Explain the recuperation test for an open well. [5]
b) Discuss the assumptions and limitations of Dupuit's theory. [5]

